Climate Conditions and Flooding Disasters in Nova Scotia

Data Analysis STAT 4620/5620 Winter 24-25

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https://github.com/KyleA2001/STAT4620_5620-Flooding_Project.git

Abstract

Summary in 150 words (Best done last, summarizes the whole project)

Keywords (max 10)

- Storm Surge
- Generalized Additive Mixed Models (GAMM)
- Time Series
- Autocorrelation Function (ACF)
- Cross-Correlation Function (CCF)

Introduction

Over the past decade, Nova Scotia has experienced an increasing number of flooding events resulting from thunderstorms, heavy rain, and extreme weather conditions. These events have significantly impacted communities, resulting in evacuations, property damage, infrastructure loss, and loss of life. The province's geographic location along the Atlantic Ocean and its coastal regions make Nova Scotia vulnerable to these events. As the frequency and severity of extreme weather conditions continue to rise due to climate change, there has been an growing interest to study how various climatic factors are related to these flooding events.

One of the most significant components involved with flooding is storm surge, the sudden rise in water level which is created by both the water level and the tide height. Storm Surge can be derived using the difference from the observed water level and the fitted tide values. In this study, the observed water level data were obtained from the Government of Canada Data Archives, while the fitted tide values were calculated using the R package "ftide." Using both these values, the storm surge response variable was derived, which is used to explore which climatic factors are most closely associated with storm surge.

Modeling these extreme weather events can be challenging due to their rarity and the complex way multiple climatic factors may interact. Although flooding events are relatively infrequent, their impact can cause Significant societal and economic damage, highlighting the need for improved forecasting tools. This research seeks to understand the relationship between various climatic factors and storm surge levels at selected locations across Nova Scotia. A secondary objective is to use these findings to forecast the likelihood of future flooding events based on the most significant contributing climatic factors.

This research will address two key questions:

1. What is the relationship between various climatic factors and storm surge levels across Nova Scotia?
2. How can the most significant correlated factors be used to predict future flooding catastrophes in Nova Scotia?

Through these investigations, this study aims to contribute valuable predictions and insights that will aid in understanding, and reducing future flooding risks in the province.

Data Description

In this section, we describe the data used for the analysis. Below is the summary table for the data obtained from the Government of Canada:

Variables	**Units**	**Description**	**Frequency**
Date	days	Date when data was recorded	Daily
Location		Halifax and Yarmouth	
Water Level	m (meters)	Height above datum or reference system	Hourly
Storm Surge	m (meters)	Water rise due to water level and tide height	Hourly
Air Temperature	°C	Air temperature	Daily min/max, Hourly avg
Wind Speed	km/h	Wind speed	Daily min/max, Hourly avg
Rainfall	mm	Precipitation (not including snow)	Real daily value

Table 1: Data description summary

The following covariates will be used in the model: air temperature, rain, and wind speed. These variables were selected because of their potential to influence storm surge levels, with each factor known to play a role in flood behavior. The goal is to fit these covariates into a model to determine whether any of them can significantly predict the response variable, storm surge.

Various covariates were considered for inclusion in the model, given the potential influence of multiple climatic factors on storm surge. However, including too many variables could result in overfitting, where the model captures noise rather than meaningful patterns, leading to inaccurate predictions. To avoid this, a simpler approach was taken by narrowing down the key variables that have the strongest observed relationships with storm surge.

One way flooding can occur is through heavy rainfall, which is one of the most significant factors in causing floods. If the amount of water from rain rises faster than it can return to the ocean or be absorbed by the soil, this leads to flooding. Given its significant impact on flood events, rainfall was chosen as a key variable in this study.

Rainfall itself is influenced by many factors, but one primary factor is air temperature. As temperatures rise, the atmosphere's ability to hold moisture increases, leading to greater rainfall amounts. Air temperature is one of the leading meteorological factors that interacts with heavy storms.[1] Therefore, it is important that air temperature is considered when examining storm surge.

Finally, wind speed was considered an important variable based because wind can increase flood magnitudes, durations, and spatial extents.[2] This highlights the significant role that wind plays in the intensity and reach of flooding events, especially along coastal regions. Given that Nova Scotia is a coastal province, the impact of wind on flood events is particularly relevant. Thus, wind speed was included in this study as a critical climatic factor to understand its contribution to storm surge and flooding, especially in combination with other factors like rainfall and air temperature.

Overall, these three covarites were decided to use in the study with storm surge due to their importance, and creating a simpler model.

Storm surge is a critical value in flood and storm impact analysis, but it cannot be directly observed. Instead, it must be calculated using the following formula:

$$\text{Storm Surge} = \text{Observed Water Level} - \text{Fitted Tide}$$

The observed water level is obtained from archived data provided by the Government of Canada. This data was measured at two separate locations: Halifax and Yarmouth.

The fitted tide uses the function ftide, where it fits the tidal data using any set of harmonic constiuents. This function takes in three main components:

- x: A numeric vector object giving the water levels; it allows for missing values.
- dto: A date vector of same length as 'x' which should be a POSIXct object; it does not allow for missing values.
- hcn: A vector of constituent names. Here, we specify only the 4 major harmonic constituents, hc4.

By applying this function to the observed water levels, the predicted storm surge values can be calculated. This function allows for the accurate estimation of storm surge at the specified locations based on tide and observations water levels.

Methods

– Provide a brief description of the analytical tools you will used – Discuss why the method is appropriate in the context of the research question and the data, including any assumptions required by the method.

[1] Pan et al. (2019)
[2] Thelen et al. (2024)

Question 1

During the initial start of the project, a various amount of methods were brought up. It was decided that the two main methods to answer question 1 of the research would be GLMs and Time Series. GLMs were considered because it would help provided a baseline to figure out what factors affect tidal storm surge. Unfortunately, there were some limitations and violated assumptions that prevented the use of this model.

Generalized Linear Models (GLMs) are a traditional linear regression that handles a wider range of data types (Continuous, count, etc…). What this model does not account for is the limitations of time-dependent data. The data collected in this research depends on the time it was collected, to find a relationship with storm surge and use for future prediction. Without the use of time data in the model, their will not be any accurate predictions.

An assumption that GLMs have is that all observations are independent. However, the data collected violates this assumption because the observations are dependent on time, meaning that values recorded at one time point are likely influenced by previous observations. This temporal dependency introduces autocorrelation, where data points closer in time are more similar, making GLMs less suitable for capturing these patterns. As a result, applying a method that accounts for time dependence was necessary to better model the data.

The dependence on time lead to the use of GAMMs (Generalized Additive Mixed Models). Generalized Additive Models (GAMs) are a versatile statistical modeling technique used to analyze complex relationships within data. Unlike linear models, GAMs can capture non-linear patterns by combining multiple smooth functions of predictor variables.[3] The model does not account for time dependence or correlated observations. This is where the model choice extends to GAMMs that accounts for both.

GAMMs assume that the random effects are independent. Since the data collected is from two different locations, the random effect from each location should be independent from each other. Another assumption is that Storm Surge should be able to follow a certain distribution. For the research the assumption is that the model follows a Normal Distribution. One of the important assumptions is temporal correlation, where observations in the past influence future values. The random effect to the model accounts for this.

ACF (Autocorrelation Function) is a method that will show how much a a variable at one time is related to its value at previous times. Since the research is studying storm surge, ACF will show if today's storm surge value was influenced from its values yesterday, before than, etc… ACF helps identity any patterns in the time series data, like seasonality or trends. If the current and past value show to be highly correlated, then the data is not random and there is some predictability in the structure.

[3] Basheer (2024)

```r
#
interpolation_df <- (read_csv(here("Data", "linear_interpolation_df.csv"),
      show_col_types = FALSE))

# Convert date to numeric format
interpolation_df$date_num <- as.numeric(
  as.Date(interpolation_df$date))

# Fit the GAMM model
halifax_gamm <- gamm(avg_water_surge ~
                       s(avg_temperature) +
                       s(avg_wind_speed) +
                       s(rain) +
                       s(date_num, bs = "cs") +
                       ti(rain, avg_wind_speed),
                       data = interpolation_df)

# Plot the ACF of residuals
acf(residuals(halifax_gamm$lme))
```

Using ACF on the GAMM model, there is high correlation between the data points, suggesting that the data collected for each time are dependent. This violates the assumption of GAMMs, and therefore makes it unusable for the model.

The other method that will be used for this analysis is Time Series. Time Series models the sequence of data points in an equally spaced time interval. The method is used to analyze trends, patterns, and behaviours over time. The importance of this model is to analyze historical data to help understand any underlying patterns, predict future values, and see how (or if any for out research) certain factors influence storm surge over time.

The goal of the research is to see what climatic factors affect storm surge over time. Time series modeling will help examine if these variables have a time-based relationship with storm surge. GAMMs was helpful for modeling non-linear, time-dependent relationships between the predictors and the response variable. However, it might not accurately represent the complex nature of time series data.

Time Series models assume that the data is stationary and no autocorrelation in the residuals. (Unsure what else to add)

Since the research analyzes storm sure data, ACF will help look at temporal dependencies. This will help identify how much past climatic factors influences storm surges behaviour.

CCF(Cross-Correlation Function) is the same as ACF, but instead examines the relationship between two different time series, with more flexibility to look at different points in time.

When using CCF for the model, it is testing how strongly the response variable (Storm Surge) is related to another variable (predictor variables like rain, wind speed, and temperature) over different lags. For the research, CCF can help to see if rain, wind speed, and/or temperature have an immediate or delayed effect on storm surge.

VAR(Vector Auto regression) (not 100% about this model)

Granger Causality (Not 100% sure)

Question 2

Since we are interested in flooding incidents in the coast of Nova Scotia, we want to identify and model the extreme storm surge levels that are associated with such events. Extreme value analysis (EVA) is a statistical method used to assess, model, and predict the behavior of extreme events or rare occurrences. It seeks to assess the probability of events that are more extreme than any previously observed. These extreme events are often characterized by their low probability of occurrence but high potential for severe consequences. One of the EVA methods utilized here is known as the Block Maxima (BM) approach. Simply, the BM analysis divides the observation period into non-overlapping periods of equal size n, known as blocks. Attention is restricted to the maximum observation in the block, known as the block maxima. In many contexts, it is common to see the block being set to a year, where our block maximas are annual maximas. These block maximas can be fitted to what is known as the Generalized Extreme Value (GEV) distribution. Given we are working with two locations, Halifax and Yarmouth, we let y_j be the vector holding the annual maximas for the j^{th} location and is of length N_j, which is the number of years observed for location j. This y_j vector contains the y_{ji} values which represent the i^{th} years annual maxima in the j^{th} location. These annual maximas are assumed to follow the GEV distribution, and have the following information:

$$y_{ji} \sim GEV(\mu_j, \sigma_j, \xi_j), \text{for } i = 1, 2, \cdots, N_j, \text{ and } j = 1, 2.$$

The density function for y_{ji} is:

$$g(y \mid \mu_j, \sigma_j, \xi_j) = \frac{1}{\sigma_j} \left\{ 1 + \xi_j \left(\frac{y - \mu_j}{\sigma_j} \right) \right\} \exp \left\{ - \left[1 + \xi_j \left(\frac{y - \mu_j}{\sigma_j} \right) \right]^{-\frac{1}{\xi_j}} \right\}.$$

Analysis

– Show the code used to actually apply the methods to the data. – Include model validation.

Bring over code

Question 2

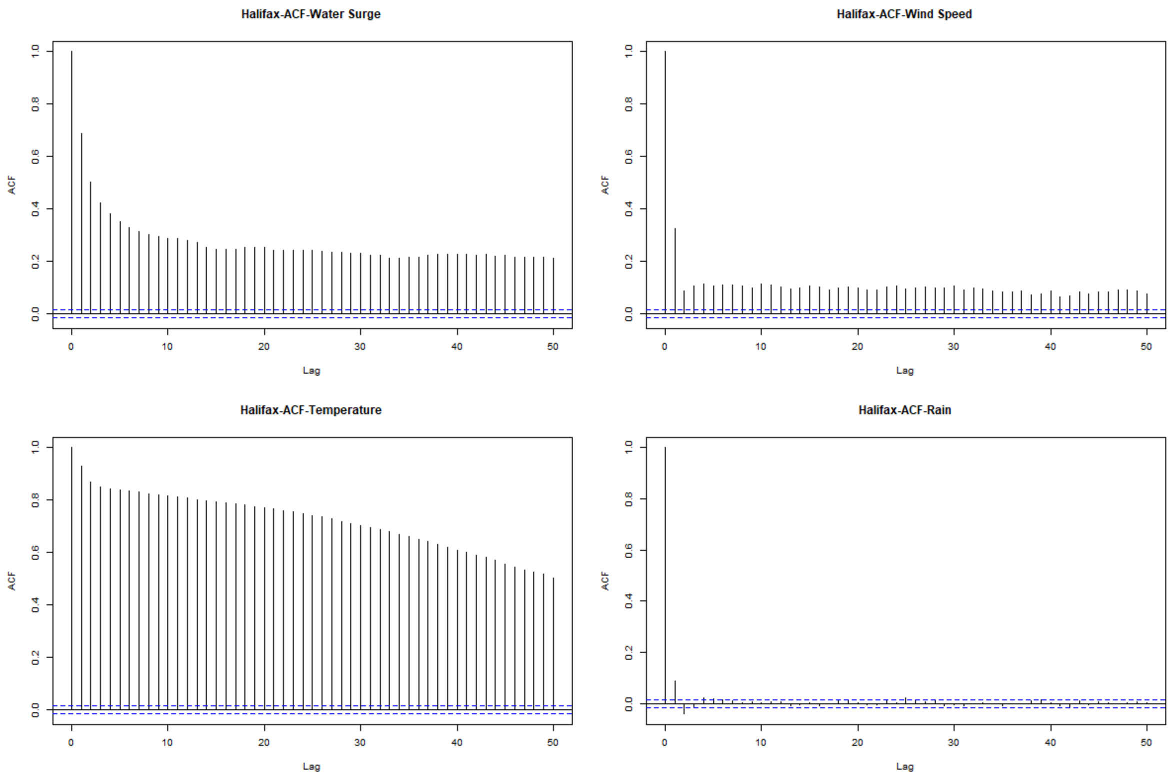

Figure 1: Autocorrelation Function (ACF) plot for Halifax for differents varirables: storm surge (top left), wind speed (top right), temperature (bottom left) and Rain (bottom right).

ACF Analysis The Autocorrelation Function (ACF) plots for Halifax and Yarmouth (Figures X and X) provide insight into how each variable correlates with itself over time. For both locations, we examine storm surge, wind speed, temperature, and rainfall. As expected, the plots display similar general patterns across locations, indicating that these variables behave in comparable ways. The ACF for storm surge, shown in the top left plot, reveals a strong temporal autocorrelation that gradually decreases but remains significant (above the blue confidence lines) even at 50 lags. This suggests that storm surge events exhibit persistent memory effects and follow cyclical patterns. In contrast, the wind speed ACF, presented in the top right plot, shows significant autocorrelation at lags 0 and 1, but the correlation rapidly declines thereafter. This suggests that wind events are relatively short-lived, with little long-term persistence. The bottom left plot displays the ACF for temperature, which exhibits the strongest and most persistent autocorrelation among all variables. The very gradual decline in correlation highlights a strong temporal persistence in temperature patterns, likely reflecting seasonal cycles and day-to-day stability. Finally, the bottom right plot presents the ACF

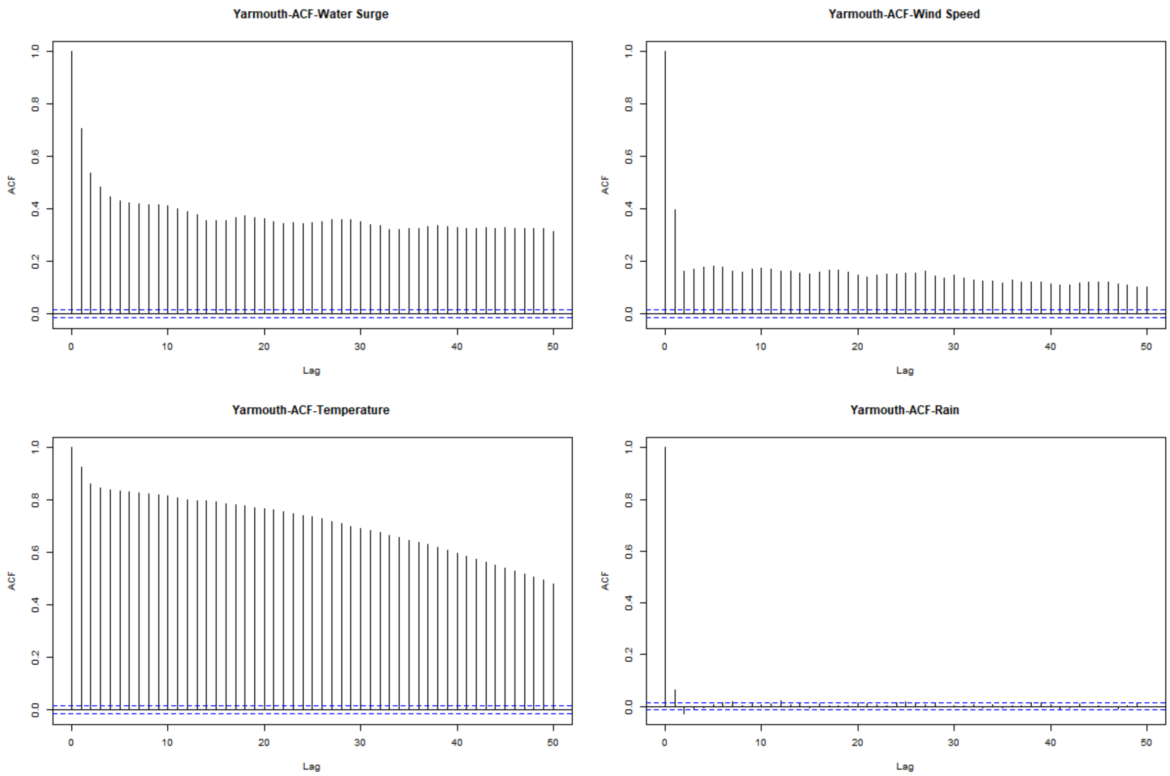

Figure 2: Autocorrelation Function (ACF) plot for Yarmourth for differents varirables: storm surge (top left), wind speed (top right), temperature (bottom left) and Rain (bottom right).

for rainfall, which shows little to no significant autocorrelation beyond lag 0. This suggests that rainfall events are largely independent from one time period to the next, with minimal influence from past occurrences. In summary, the ACF analysis highlights distinct temporal behaviors across variables. Storm surges exhibit long-term dependencies and cyclical patterns, wind speed fluctuations are short-lived, temperature is highly persistent due to seasonal effects, and rainfall events occur independently with no significant memory. These differences reflect the underlying physical and meteorological drivers governing each variable.

Cross-Correlation Analysis The Partial Cross-Correlation Function (P-CCF) plots reveal relationships between water surge and other variables. The Figure X and Figure~4 discuss respectively of the results Halifax and Yarmouth. The top plot discuss about water surge vs. wind speed, present a positive correlation at lag 0 and several subsequent lags. This suggests wind events immediately impact water surge, with effects persisting through multiple time periods. The pattern indicates wind speed is a driver of water surge events. In contrast storm surge vs. temperature shows a strong negative correlation that gradually diminishes over time. This inverse relationship is particularly pronounced for Halifax. That suggests lower temperatures are associated with higher water surge events (possibly related to winter storm patterns). The last plot on the bottom discuss about water surge vs. rain. A positive correlation at very short lags (0-1) is identifiable. The relationship weakens quickly over time, indicating that rainfall has a short-term impact on water surges but little long-term influence The patterns are generally similar between Halifax and Yarmouth, suggesting regional consistency in meteorological relationships. The temperature-surge negative correlation appears particularly strong in Halifax. Moreover, Halifax shows some interesting phase shifts in the wind-surge relationship that might indicate more complex coastal geography effects. These relationships suggest that water surge events in Nova Scotia are primarily driven by wind conditions, with secondary influences from temperature patterns and immediate rainfall. The negative correlation with temperature likely reflects seasonal storm patterns, with more significant surge events occurring during colder months

Var The analysis employed a Vector Autoregression (VAR) model to examine the relationship between multiple climatic variables and water surge dynamics over time. Prior to model estimation, the stationarity of all time series variables—average water surge, average temperature, average wind speed, and rainfall—was assessed using the Augmented Dickey-Fuller (ADF) test. The results confirmed that all variables were stationary (p-value = 0.01 for each variable), allowing for VAR modeling without the need for differencing. Different lag orders were explored, with a maximum of 20 lags considered. The selection criteria provided varying recommendations: the Akaike Information Criterion (AIC) suggested 20 lags, the Hannan-Quinn Criterion (HQ) recommended 19, while the Schwarz Criterion (SC) indicated an optimal lag order of 11. The Final Prediction Error (FPE) aligned with AIC, selecting 20 lags. This variation aligns with the well-documented tendency of AIC to favor higher lag orders due to its relatively weaker penalty on model complexity. The primary model incorporated all climatic factors using a VAR(7) specification, including average water surge, temperature, wind speed, and rainfall with seven lags. This model explained approximately 62.6% of the variation in water surge (R-squared = 0.6262) and exhibited the lowest AIC (-39278.16), suggesting the best overall fit. Among the most significant predictors, the strongest influence was the previous water surge at lag 1 (coefficient = 0.635). Wind speed showed a negative relationship at lag 1 (-0.00168) but a positive effect at lag 2 (0.00109). Rainfall displayed a delayed negative effect at lag 2 (-0.00085). Residual analysis using the autocorrelation function (ACF) plot indicated no significant autocorrelation, confirming that the model effectively captured temporal dependencies. A simplified model was then considered, incorporating only water surge and wind speed with a single lag (VAR(1)). This model explained 59% of the variance (R-squared = 0.5908) and had a higher AIC (-37666.21), suggesting a comparatively poorer fit. However, both included variables remained highly significant ($p < 2e-16$), with wind speed showing a consistent positive effect (0.000702) on water surge. Although residual autocorrelation was minimal, it was slightly higher than in the full VAR(7) model. To further isolate the influence of past water surge levels, an autoregressive (AR) model was estimated. This model, focusing exclusively on the temporal dependence of water surge, had an R-squared of 0.4713, explaining 47% of the variance. The AIC value was -38196.53, positioning it between the simplified VAR(1) model and the full VAR(7) model. The autoregressive coefficient (0.6865) indicated a strong temporal dependence, reinforcing the role of past water surge levels as the primary determinant of future surges. The results highlight key insights into the dynamics of water surge variation. Water surge exhibits strong autocorrelation, with previous surge levels serving as the most influential predictor. The inclusion of climatic factors significantly improves predictive power, increasing explained variance from 47% (AR model) to 62% (full VAR model). Wind speed exerts both immediate and delayed effects, displaying an initial negative influence followed by positive effects at later lags, suggesting a complex interaction with water dynamics. Rainfall primarily exerts a delayed negative impact, with significant effects observed at lags 2, 3, and 7, whereas temperature influences are less consistent across time lags. Overall, while the full VAR(7) model provides the best fit for accurately capturing storm surge dynamics, the VAR(1) model with wind speed represents a viable alternative for operational use, balancing predictive capability with model simplicity. The absence of significant overfitting, as confirmed by time series cross-validation, further supports the robustness of these models. These findings underscore the complex interplay between past water surge levels and climatic factors in shaping surge dynamics in Halifax, emphasizing the importance of multivariate approaches for improved forecasting.

GAMM The analysis employed a Generalized Additive Mixed Model (GAMM) to investigate the relationship between water surge and climatic predictors, accounting for location as a random effect. The model incorporated smooth terms for temperature, wind speed, rainfall, and date, allowing for the capture of seasonal patterns and nonlinear trends in water surge dynamics. The autocorrelation function (ACF) plots reveal a strong temporal structure in the data. The random effect component exhibits a high autocorrelation at lag 1, around 1, which gradually decreases but remains above the significance threshold after 45 lags. This pattern suggests that the random effect structure does not fully account for the temporal dependencies present in the data. Similarly, the smooth component demonstrates a comparable autocorrelation at lag 1 but with a more persistent decline. The fact that autocorrelation remains significant across all lags indicates that residual temporal dependencies remain unmodeled within the smooth terms. Residual diagnostic plots provide further insight into the model's performance. The residuals versus fitted values plot for the GAM shows a fairly uniform spread around zero, though some deviation is observed, particularly at higher fitted values where extreme residuals become more prominent. The quantile-quantile (QQ) plot highlights moderate departures from normality, particularly in the upper tail, where observed values exceed theoretical expectations. The residuals from the linear mixed effects (LME) component display a similar pattern, albeit with slightly improved normality. Nevertheless, minor deviations from normality persist at both extremes, reinforcing the idea that some residual structure remains unaccounted for in the model. The findings underscore the presence of persistent temporal autocorrelation, indicating that the model may benefit from refinements. The inclusion of an explicit autocorrelation structure, such as a first-order autoregressive (AR(1)) correlation term, could help in capturing the remaining dependencies. Additionally, incorporating more time-related predictors or interactions may enhance explanatory power. Alternative smoothing parameter adjustments or different basis functions could refine the model's ability to account for temporal trends. Despite these areas for improvement, the model effectively captures major patterns in the data. Residuals are generally well-behaved, displaying symmetry around zero and no clear signs of heteroscedasticity. While slight deviations from normality exist, they do not pose substantial issues for inference. The model offers a strong foundation for understanding water surge dynamics, but adjustments to better account for temporal dependencies could further enhance its predictive performance and overall robustness.

Question 2

EVA Looking at the overall pattern, both locations show storm surge return levels increasing with longer return periods, following a logarithmic curve that flattens at higher return periods. This is typical in extreme value analysis with the difference between a 50-year and 100-year event is less dramatic than between a 2-year and 10-year event. Comparing both location, Halifax consistently shows higher return levels (approximately 0.08-0.10m higher) compared to Yarmouth across all return periods. Halifax's 100-year return level reaches about 0.71m while Yarmouth's reaches about 0.62m. The Figure X and X represent the case with wind as covariates. The covariate analysis produced remarkably similar results to the initial analysis. For Halifax, differences between initial and covariate models are extremely small (0.0001-0.0003m), for Yarmouth, differences are slightly larger but still minor (0.0049-0.0116m). Both graphs show upper and lower confidence bounds (red dotted lines). The confidence interval widens as the return period increases, reflecting greater uncertainty in predicting more extreme events. The minimal difference between the initial and covariate models suggests that wind speed doesn't significantly improve storm surge prediction beyond what the initial EVA model already captures. This could mean that wind speed is already implicitly represented in your storm surge data; the relationship between wind speed and storm surge in these locations isn't strong enough to substantially improve predictions or, that other factors might be more important in determining extreme storm surge events

Results

– Present and discuss the outputs of your methods. – Include any visualizations of the analysis.

Bring over code

Conclusions

– Discuss the results of the previous section in the context of the research question. – Give an answer to your research question and discuss any uncertainty in your results. – If appropriate, discuss why the research question cannot be fully answered using the data and techniques available.

Missingness The presence of missing values in our variable of interest could pose a challenge in this project. Many of the time series methods used in this report typically do not support missing data. However, our dataset contains some time gaps. For the first question, we only included covariate data when the response variable was available, which led to the removal of some information from our dataset. Missingness can be problematic as it results in the loss of valuable data. To address this, we used a time interval without gaps, which is shorter than the initial interval but still assumed to be sufficiently long to provide a realistic representation of the dataset. Then for the second question, we considered methods for predicting missing values to mitigate this issue.

Covariates For the second question, we aimed to predict storm surges, initially using only water surge data and then incorporating additional correlated covariates. We expected different results, potentially leading to more accurate predictions. However, adding covariates did not significantly improve the predictions, suggesting that storm surge is primarily influenced by storm surge itself. This raises several discussion points regarding the model: - A flawed implementation can yield unsatisfactory results. Even if we apply the formula as suggested in R, the literature indicates that incorporating covariates requires a more complex approach than initially assumed. - We selected only highly correlated covariates for the model. However, adding a strongly correlated variable may not provide additional useful information. The literature suggests that experimenting with less correlated variables could lead to more meaningful improvements.

For future analysis, we might consider testing different covariates (atmospheric pressure, tide levels, storm track). We can also examining if the covariate relationship changes seasonally. An other option would be to investigate if there are specific storm types where wind speed becomes more significant. As an alternative if none of this proposition is satisfying, predicting water surge levels could be approached using a simpler model. Instead of employing Extreme Value Analysis (EVA) for long-term predictions with covariates (a complex procedure), we propose using Long Short-Term Memory (LSTM) methods to make short-term predictions, leveraging the results obtained in the first question.

Non stationarity A critical aspect of the model that needs to be highlighted is its non-stationarity in the context of EVA. Our response variable, storm surge, is influenced by sea level, which has been shown to be increasing over time (as documented in the literature). Consequently, storm surge is also expected to rise over time. Despite this, we assumed that non-stationarity does not affect our results, even though this assumption might not hold. Further investigation may be required to assess the impact of this assumption on our findings.

References

– Cite data sources. –

Cite literature and other sources useful for explaining the background information, research question, and analytical tools.

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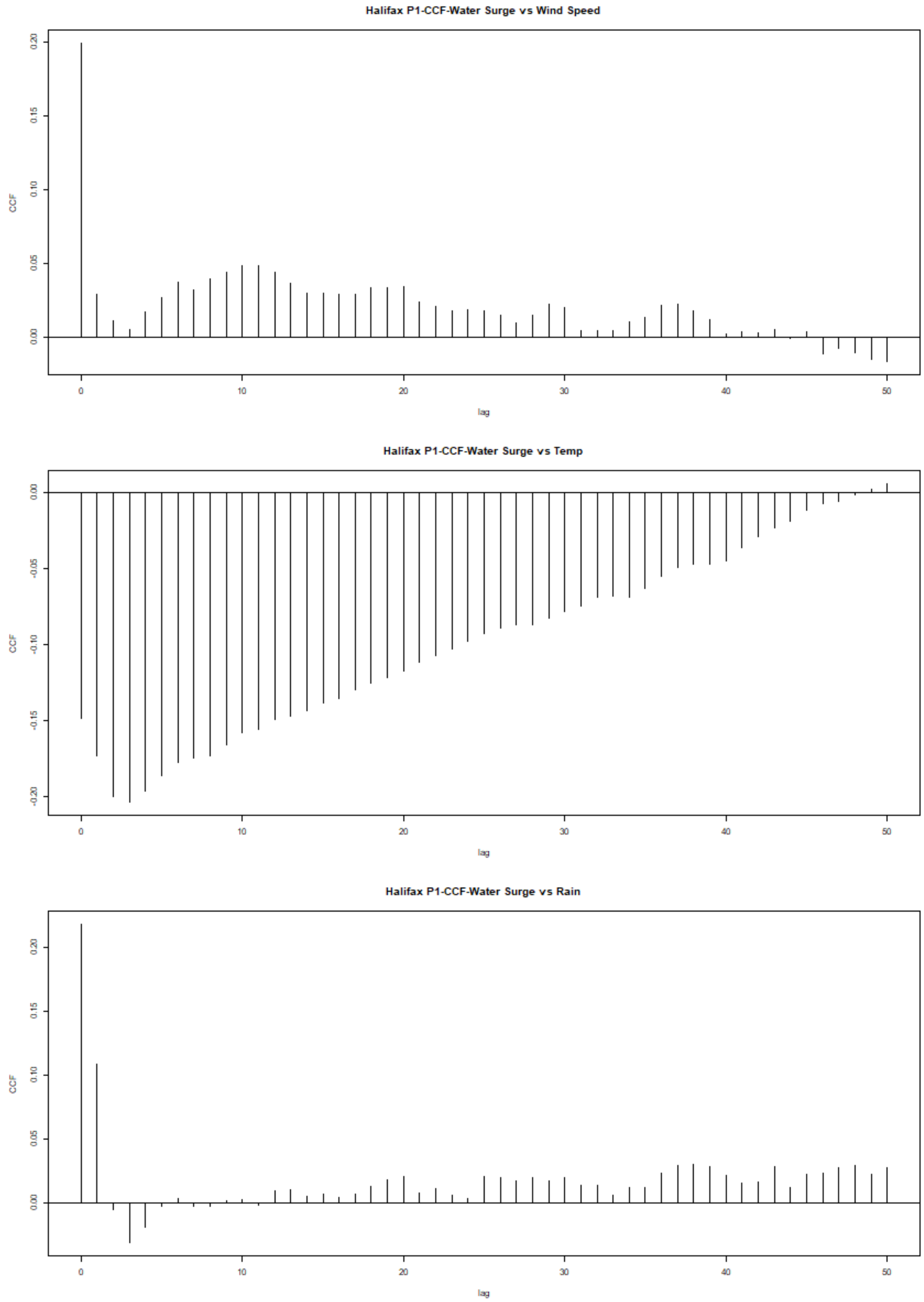

Figure 3: Cross correlation Function (CCF) plot for Halifax considering differents variables: storm surge vs wind speed (top), storm surge vs temperature (middle), storm surge vs rain (bottom).

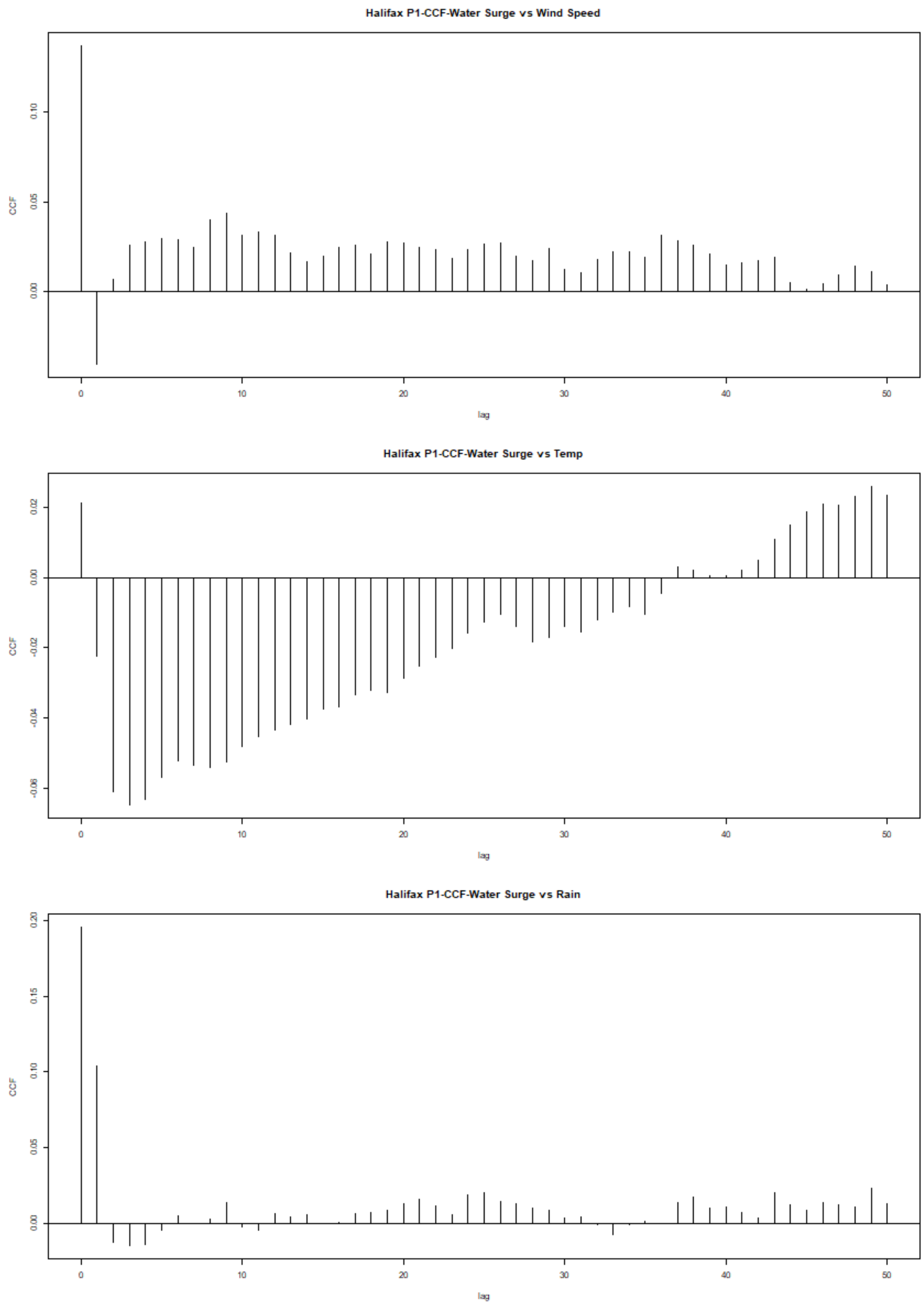

Figure 4: Cross Correlation Function (CCF) plot for Yarmourth considering differents variables: storm surge vs wind speed (top), storm surge vs temperature (middle), storm surge vs rain (bottom).